

VARUN BHADURGATTE NAGARAJ

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EDUCATION

Round Rock High School

Cedar Park, TX · Graduation: May 2027
STEM & Business Industry Academy
4.0 GPA · SAT: 1490 | Math: 780, English: 710

Coursework

AP Calculus BC, AP CS A, AP Physics I, AP Research, Digital Electronics, AP Seminar

TECHNICAL SKILLS

Languages

Python, Java, C/C++, JavaScript, TypeScript, HTML/CSS, R, SQL

Web Development

React, React Native, Flask, Supabase (Postgres), Redis, Vercel, Qiskit

AI Development

ML/NLP, LLM Integration, MCP Agents, Vector Databases, RAG, Ollama, Inference Optimization

Quantum Mechanics

Qiskit, IBM Quantum, AerSimulator, Contextual Circuits, ZNE, Noise Modeling

Robotics

PROS, LemLib, AprilTag Vision, Arduino, Autodesk Inventor, Java, PID Control, VEXcode, AutoCAD, Auto-Pathing

CERTIFICATIONS

Business Management & Administration
NOCTI · Apr 2025

Autodesk Certified: Inventor
Certiport · Apr 2024

MOS: Word, Excel, PowerPoint
Certiport · May 2023 · Office 2019

Leadership & Activities

Boy Scouts of America

Life Scout · 2020 – Present

- Built and deployed a web app for trail signage (QR-accessed) that identifies species from user-submitted images using an AI model and provides habitat information (Eagle Scout project).
- Engineered an embedded flood detection system (Arduino, ultrasonic sensing) for low-lying bridges, enabling real-time alerts to improve local safety.
- Led and mentored scouts in advancement, first aid, and emergency preparedness.

Austin Kannada Sangha (AKS)

Tech Volunteer · 2020 – Present

- Volunteer for 2,000+ attendee festivals: supported ticketing, payments, website updates, and backstage logistics.

SEWA – Meals for Students

Volunteer · 2024

National Honor Society

Member · Oct 2025 – Present

WORK EXPERIENCE

Round Rock High School Co-Op

Mar 2025 – Present

Chief Executive Officer

- Directed a 250+ member, multi-department enterprise; set OKRs, led cross-functional execution, and secured district approval to formalize CO-OP as a credited CTE program.
- Scaled e-commerce operations to 15,000+ monthly views; partnered with district staff and educators to design and implement inclusive fulfillment workflows, establishing procedures that expanded roles for FAC and disabled students.
- Modernized processes through automation and applied ML: scheduling, demand forecasting, performance analytics, and computer vision-based inventory tracking.

HallHop – Digital Pass System

Nov 2024 – Present

Founder & Lead Developer

- Built and deployed a full-stack digital hall pass platform (Chrome extension/web app + backend) replacing paper workflows.
- Developed custom data integration and REST API syncing with district grade systems (HAC); adopted by partner student organizations, handling 4,000+ external requests per month.
- Engineered FERPA-compliant, high-performance infrastructure with multi-tenant data isolation and optimized asynchronous pipelines, achieving 7× performance improvement.
- Developing indoor positioning system using Wi-Fi-based triangulation for real-time student tracking and emergency response (2–4m accuracy).

Virtual Robot Simulator (VRS)

May 2025 – Present

Project Team Lead

- Led VEX integration for a cross-platform robotics simulation system, extending tooling originally built for FTC to support additional hardware and control workflows.
- Led development of an in-house FTC control SDK, designing motor control abstractions and enabling simulation integration; contributed to scaling the platform to 32,000+ monthly users.
- Improved Java to JavaScript transpilation pipeline for FTC SDK code, increasing performance and correctness of Unity-based robot simulation.
- Mentored FTC teams in autonomous systems and computer vision, focusing on Java control systems and AprilTag-based localization.

Transforze

Aug 2025 – Present

Software Engineer

- Developed a multi-tenant CRM platform with real-time agent workflows and secure data isolation across workspaces.
- Built agent-driven automation systems integrating with productivity tools (Gmail, Calendar, Drive), enabling task execution through conversational interfaces.
- Implemented end-to-end ML pipeline for conversational AI features, including dataset curation, intent taxonomy design, model training, and evaluation (relevance, accuracy, latency).

RESEARCH

AP Research · Round Rock HS

Aug 2025 – May 2026

Evaluating Quantum Contextuality as an Error Mitigation Resource in the Mermin–Peres Magic Square Game

- Designed and executed experiments on IBM Quantum hardware (8,192 shots × 9 configurations) to evaluate superconducting qubit performance.
- Implemented zero-noise extrapolation (gate folding, Richardson extrapolation), improving fidelity from 0.753 to 0.805.

Texas State University

May 2025 – Jan 2026

Research Contributor

- Co-authored and built R-based analysis pipeline using SEM and multiple imputation on a 600+ student survey to study composting behavior (Theory of Planned Behavior).

ROBOTICS

VEX Robotics Competition (VRC)

Oct 2024 – Present

Lead Programmer — Virus VRC | State Championship Qualifier

- Designed and tuned a Ziegler–Nichols PID controller; implemented odometry via IMU, tracking wheel, and encoder fusion; achieving 5× improvement in autonomous consistency with a 90%+ success rate.

First Tech Challenge (FTC) Robotics

Jan 2023 – Dec 2024

Hardware Team Lead

- Designed and integrated competition drivetrain and full robot system; developed Java-based autonomous and teleop control with AprilTag vision, ensuring reliability and durability under match conditions.